FSD DOCUMENTATION FORMAT

1. Introduction

Project Title: Strategic Product Placement Analysis (SPPA)

Team Members:

- Golla Mahendra Babu – Team Lead (Data Collection, Data Preparation, Visualizations, Dashboard, Web Integration, Testing)
- Seereddy Harshitha Reddy – Team Member (Story Development)

2. Project Overview

Purpose: To enable retail managers to make data-driven decisions regarding product placement, pricing strategies, and promotional effectiveness through interactive Tableau dashboards and stories.

Key Features:

- 8 interactive visualizations (Bar Chart, Stacked Bar Chart, Donut Chart, Scatter Plot, Line Chart, Pie Chart, Bubble Chart, Heat Map)
- Interactive dashboard with dynamic filtering
- Narrative-driven Tableau Story
- Responsive Bootstrap website
- Tableau Public integration

3. Architecture

Frontend: HTML5, CSS3, Bootstrap 5; Embedded Tableau visualizations via iframe/JavaScript API

Backend: Tableau Public (Cloud-based visualization hosting); No custom backend required

Database: Tableau Data Extracts from CSV source; Local CSV file storage

4. Setup Instructions

Prerequisites: Tableau Desktop (for development), Tableau Public account (for publishing), Visual Studio Code (for web development), Modern web browser, Git.

Installation:

- Clone repository: git clone [https://github.com/mahi-27104/Strategic-Product-Placement-Analysis]
- Open project folder in Visual Studio Code
- Open index.html in browser to view website
- For Tableau development: Open Tableau Desktop and import Dataset.csv

5. Folder Structure

```
SPPA-Project/
├── index.html             # Main HTML file
├── style.css              # CSS styles
├── Dataset.csv            # Retail dataset
├── README.md              # Project documentation
└── assets/
    └── images/            # Screenshots and assets
```

6. Running the Application

Local Website: Open index.html directly in browser or use Live Server extension in VS Code.

Tableau Development: Open Tableau Desktop, Import Dataset.csv, Create visualizations, Publish to Tableau Public.

7. API Documentation

N/A – No custom APIs. Uses Tableau Public for embedding.

8. Authentication

N/A – Public dashboard accessible without authentication.

9. User Interface

Screenshots: Hero section with project title, About section with key features, Dashboard section with embedded Tableau dashboard, Story section with embedded Tableau story, Team section with project members.

10. Testing

- Functional testing of all visualizations
- Performance testing for load time
- Cross-browser compatibility testing
- Responsive design testing

11. Screenshots / Demo

Tableau Dashboard: [public link] | Video Demo: [Google Drive Link]

12. Known Issues

- Tableau Public may have occasional loading delays depending on network
- Limited to dataset size (1,000 records)

13. Future Enhancements

- Integration with real-time sales data
- Predictive analytics for sales forecasting
- Additional visualization types
- User authentication for personalized views
- Export functionality for reports
- Integration with additional data sources

FINAL REPORT TEMPLATE

1. INTRODUCTION

1.1 Project Overview

Strategic Product Placement Analysis (SPPA) is a Business Intelligence project that leverages Tableau to analyze retail sales data. The project investigates how pricing, promotions, competitor pricing, product positioning, and seasonal demand influence sales. The complete BI workflow is demonstrated—from data collection and preprocessing to visualization, dashboard creation, story development, publishing, and website integration.

1.2 Purpose

To provide retail managers and stakeholders with interactive, data-driven insights for optimizing product placement, pricing strategies, and promotional effectiveness. The project aims to transform raw retail data into actionable intelligence that supports strategic decision-making.

2. IDEATION PHASE

2.1 Problem Statement: Retail stores face challenges in determining optimal shelf positions and pricing strategies to maximize sales. This project addresses these challenges by leveraging Tableau to build interactive dashboards. (See Section 2/3 for detailed Empathy Map and Brainstorming).

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map: See Section 4. | 3.2 Solution Requirements: See Section 5. | 3.3 Data Flow Diagram: See Section 6. | 3.4 Technology Stack: See Section 7.

4. PROJECT DESIGN

4.1 Problem-Solution Fit: See Section 8. | 4.2 Proposed Solution: See Section 9. | 4.3 Solution Architecture: See Section 10.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning: See Section 11.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Functional Testing: See Section 12. | 6.2 Performance Testing: See Section 12. | 6.3 UAT Results: See Section 13.

7. RESULTS

7.1 Output Screenshots

Dashboard: Interactive dashboard with 8 visualizations, Filter controls for Product Category/Promotion Status, Layout containers for structured display.

Story: Sequential analysis narrative, Captions and annotations for each story point, Logical progression through pricing/sales/demographics.

Website: Hero section, about section with features, Dashboard/Story embed sections, Team section.

8. ADVANTAGES & DISADVANTAGES

#	Advantage	#	Disadvantage
1	Interactive dashboard enables real-time data exploration	1	Limited to Tableau Public's free tier capabilities
2	8 visualization types provide comprehensive analysis	2	Dataset size limited (~1,000 records)
3	Dynamic filtering allows customized views	3	No real-time data integration
4	Responsive website ensures accessibility	4	Requires internet connection for Tableau visuals
5	Tableau Story provides narrative-driven insights	5	Limited customization for advanced analytics
6	Publicly accessible via Tableau Public		
7	Low-cost solution using free tools		

9. CONCLUSION

This project successfully demonstrates the complete Business Intelligence lifecycle using Tableau. The raw retail data was effectively transformed into meaningful insights that support informed decision-making regarding product placement and pricing strategies. The project proves that BI tools can transform complex data into actionable intelligence for strategic planning.

10. FUTURE SCOPE

#	Future Enhancement	Description
1	Real-time Data Integration	Connect to live sales data sources
2	Predictive Analytics	Add forecasting models for sales prediction
3	Additional Visualizations	Incorporate more chart types for deeper analysis
4	User Authentication	Add login for personalized dashboards
5	Export Functionality	Enable PDF/Excel export of reports
6	Mobile App	Build dedicated mobile application
7	ML Integration	Add machine learning for pattern recognition
8	Multi-store Comparison	Support for multiple store locations

11. APPENDIX

Dataset: Kaggle Retail Dataset | Repo: GitHub | Demo: Video Link | Dashboard/Website: Dasboard

Performance Test Templates (For Tableau Project)

Tableau Performance Test Template

S.No	Parameter	Screenshot / Values
1	Data Rendered	All 1,000 records successfully rendered in Tableau
2	Data Preprocessing	Missing values handled; calculated fields created; data types validated
3		

	Utilization of Filters	Interactive filters applied: Product Category, Promotion Status, Customer Demographics, Seasonal Demand
4	Calculated Fields Used	Price Difference, Promotion Status, Discount Category, Sales per Price
5	Dashboard Design	8 Visualizations: Bar, Stacked Bar, Donut, Scatter, Line, Pie, Bubble, Heat Map
6	Story Design	Sequential story with captions and annotations covering pricing analysis, sales trends, customer demographics, and promotional effectiveness